

Qwen Cosine Similarity Between Kept Steering Vectors

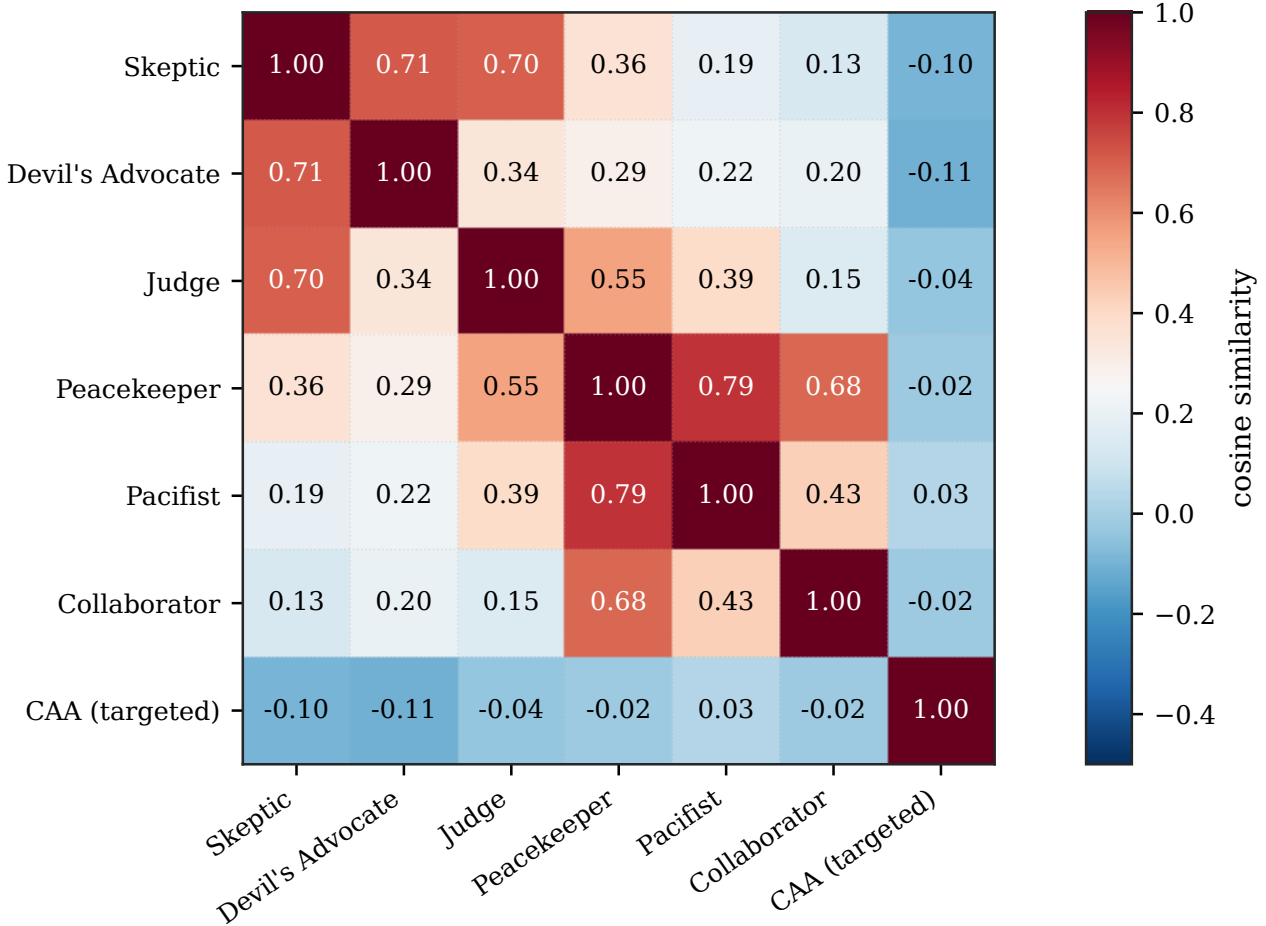


Figure 5. Cosine similarity heatmap. Critical roles cluster ($\cos \approx 0.6$ – 0.7); conformist roles cluster separately ($\cos \approx 0.8$). All role–CAA cosines are < 0.17 , but *signs* differ across models (see text).

- No capability side-effect evaluation.** We do not test whether steering affects general capabilities (e.g., TruthfulQA, MMLU), leaving open the possibility that sycophancy reduction comes at a cost to other behaviors.

G. Reproducibility

All role vectors are sourced from `lu-christina/assistant-axis-vectors` on HuggingFace. CAA vectors are extracted following Rimsky et al. (2024) from disjoint datasets (`nlp_survey` and `political_typology`). Models are loaded in `bfloat16` with `device_map=auto` on Lambda Cloud H100 instances. Experimental pipeline (data processing, steering, evaluation, statistical analysis): <https://anonymous.4open.science/r/Sycophancy-Steering-9DF0/>. Results, figures, and analysis notebooks: <https://anonymous.4open.science/r/sycophancy-clean-results-585C>.

H. Use of AI Assistants

Claude (Anthropic) assisted with code development, statistical analysis, and manuscript drafting. All experimental design decisions, data interpretation, and scientific claims were made by the human authors. No AI system generated or modified experimental data.

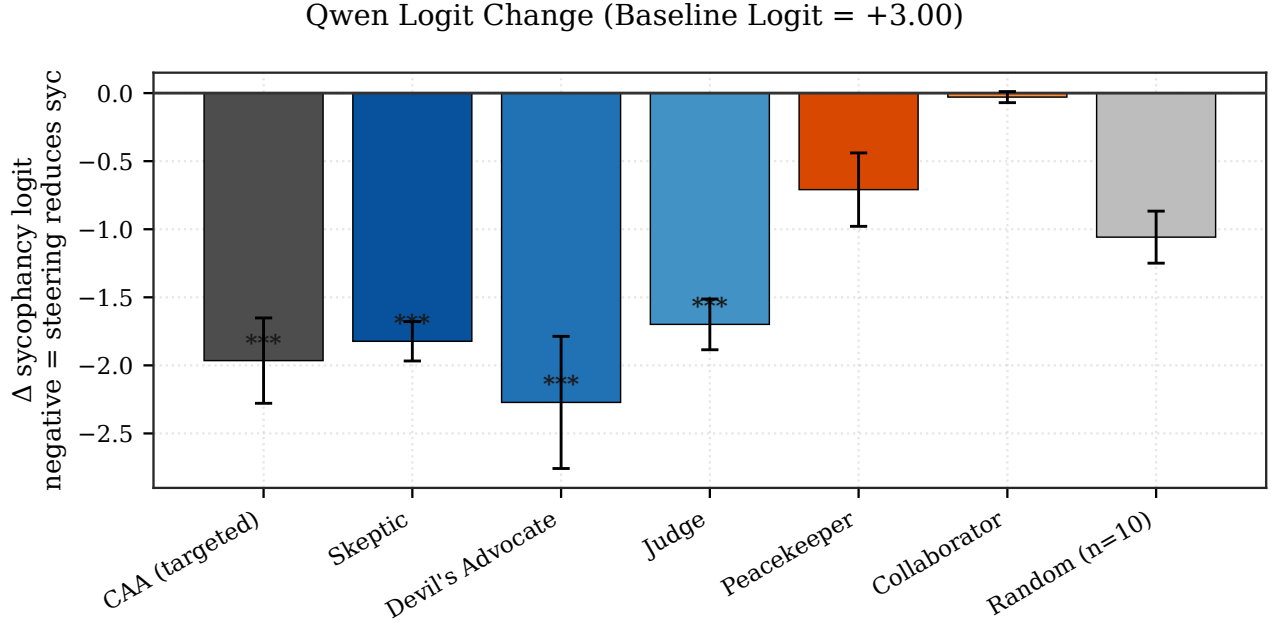


Figure 6. Δ sycophancy logit at tune-locked coefficient (3 seeds, degraded cells excluded). Error bars show 95% CIs; — = Holm-significant on all 3 seeds.

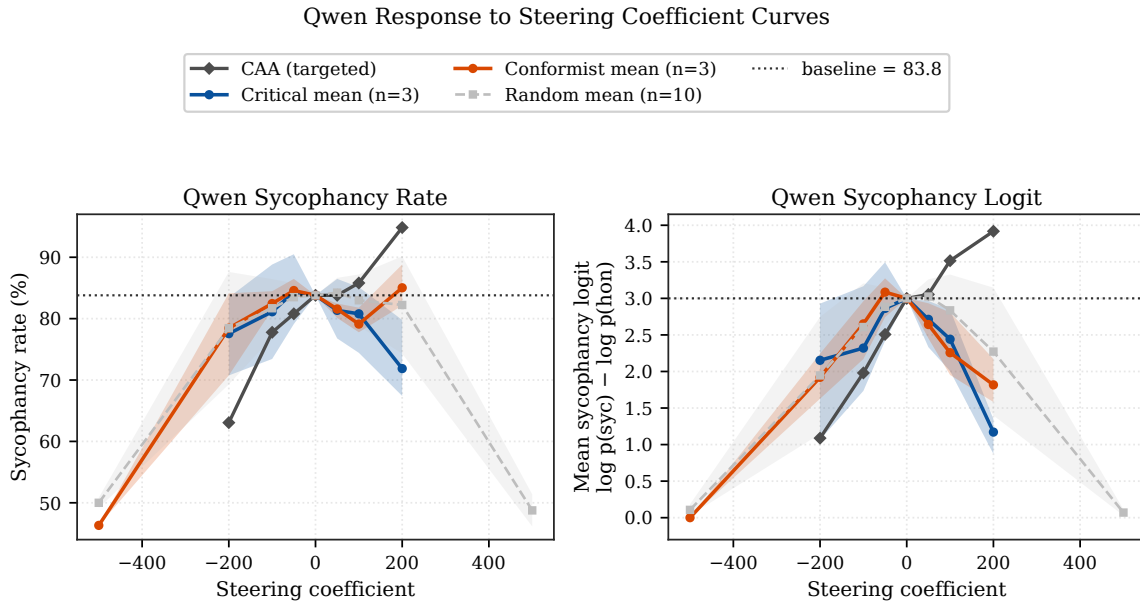


Figure 7. Per-condition steering curves (kept conditions only). Rows = metric; columns = model; shaded bands = random-control mean $\pm$ std. Each line is a single condition; degraded cells excluded.

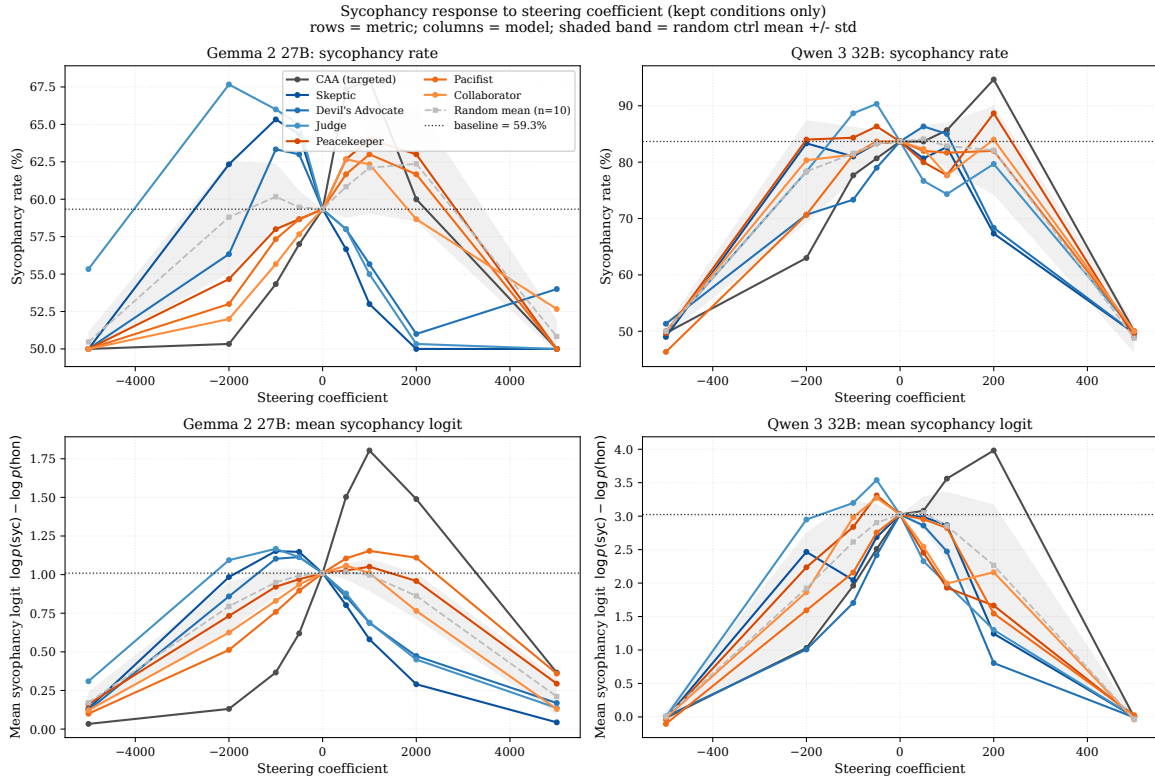


Figure 8. Per-condition steering curves (kept conditions only). Rows = metric; columns = model; shaded bands = random-control mean $\pm$ std. Each line is a single condition; degraded cells excluded.